

References

1. Haghjoo, A., Hemmati, S., Mobasher, B., Chartab, N., de la Vega, A., Eifler, T., Ramey, E., Nayyeri, H., & Sattari, Z. (2026). Learning to See Sharper: A Physics-Informed Artificial Intelligence Framework for Super-Resolving Galaxy Spectra. *The Astrophysical Journal*, submitted.
2. Eisenstein, D. J., Willott, C., Alberts, S., et al. (2026). Overview of the JWST Advanced Deep Extragalactic Survey (JADES). *The Astrophysical Journal Supplement Series*, 283(1), 6.
3. Gardner, J. P., Mather, J. C., Abbott, R., et al. (2023). The James Webb Space Telescope Mission. *Publications of the Astronomical Society of the Pacific*, 135(1048), 068001.
4. Jakobsen, P., Ferruit, P., Alves de Oliveira, C., et al. (2022). The Near-Infrared Spectrograph (NIRSpec) on the James Webb Space Telescope. I. Overview of the instrument and its capabilities. *Astronomy & Astrophysics*, 661, A80.
5. Curtis-Lake, E., Carniani, S., Cameron, A., et al. (2023). Spectroscopic confirmation of four metal-poor galaxies at $z = 10.3\text{--}13.2$. *Nature Astronomy*, 7, 622–632.
6. Ledig, C., Theis, L., Huszár, F., et al. (2017). Photo-Realistic Single Image Super-Resolution Using a Generative Adversarial Network. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* (pp. 4681–4690).
7. Vaswani, A., Shazeer, N., Parmar, N., et al. (2017). Attention Is All You Need. In *Advances in Neural Information Processing Systems*, 30.
8. He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep Residual Learning for Image Recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* (pp. 770–778).
9. Schawinski, K., Zhang, C., Zhang, H., Fowler, L., & Santhanam, G. K. (2017). Generative adversarial networks recover features in astrophysical images of galaxies beyond the deconvolution limit. *Monthly Notices of the Royal Astronomical Society: Letters*, 467(1), L110–L114.
10. Fabbro, S., Venn, K. A., O’Brian, T., et al. (2018). An application of deep learning in the analysis of stellar spectra. *Monthly Notices of the Royal Astronomical Society*, 475(3), 2978–2993.
11. Lucy, L. B. (1974). An iterative technique for the rectification of observed distributions. *The Astronomical Journal*, 79, 745.
12. Richardson, W. H. (1972). Bayesian-Based Iterative Method of Image Restoration. *Journal of the Optical Society of America*, 62(1), 55–59.
13. Tikhonov, A. N. (1963). Solution of Incorrectly Formulated Problems and the Regularization Method. *Soviet Mathematics Doklady*, 4, 1035–1038.
14. Rudin, L. I., Osher, S., & Fatemi, E. (1992). Nonlinear Total Variation Based Noise Removal Algorithms. *Physica D: Nonlinear Phenomena*, 60(1–4), 259–268.

15. Scourfield, M., Saintonge, A., de Mijolla, D., & Viti, S. (2023). De-noising of galaxy optical spectra with autoencoders. *Monthly Notices of the Royal Astronomical Society*, 526, 3037–3050.
16. Kennicutt, R. C. (1998). Star Formation in Galaxies Along the Hubble Sequence. *Annual Review of Astronomy and Astrophysics*, 36, 189–232.
17. Kewley, L. J., Dopita, M. A., Sutherland, R. S., Heisler, C. A., & Trevena, J. (2001). Theoretical Modeling of Starburst Galaxies. *The Astrophysical Journal*, 556, 121–140.